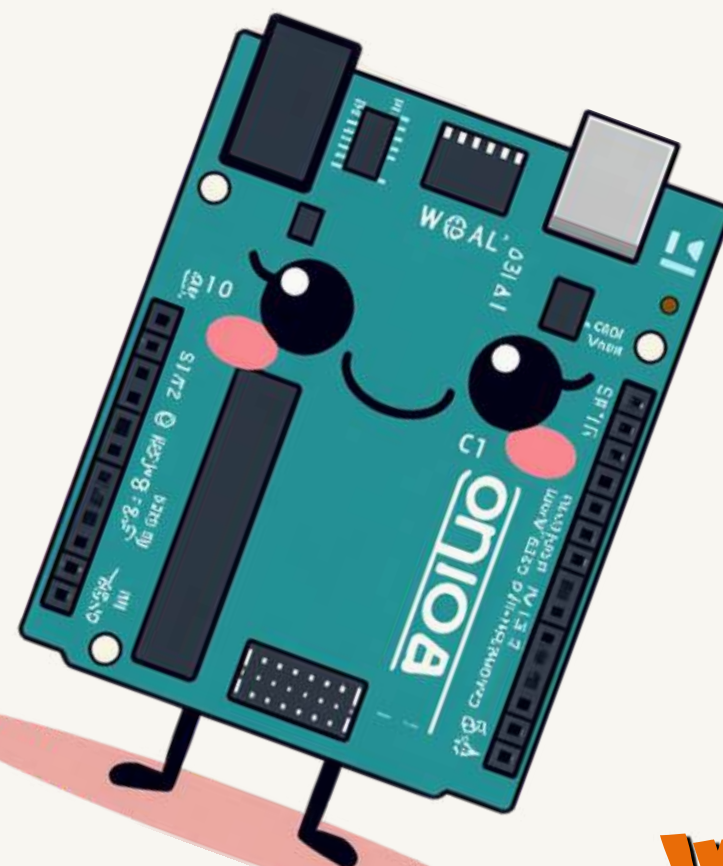


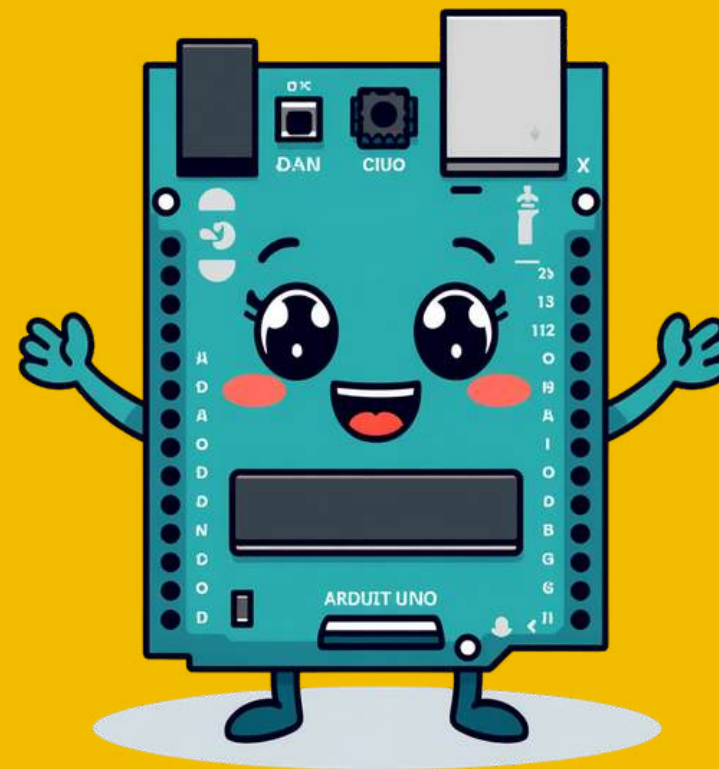


SPINNING WHEELS & DRIVING MOTORS! WITH ARDUINO AND L298N

HOW DOES IT MOVE FORWARD AND BACKWARD?



MEET YOUR ROBOT'S BRAIN & MUSCLES



THE BRAIN

Arduino Uno is the microcontroller — it thinks, decides, and sends instructions to all parts of the robot.

THE MUSCLES

The DC Motor is the muscle! It spins the wheels and makes your robot actually move forward or backward.

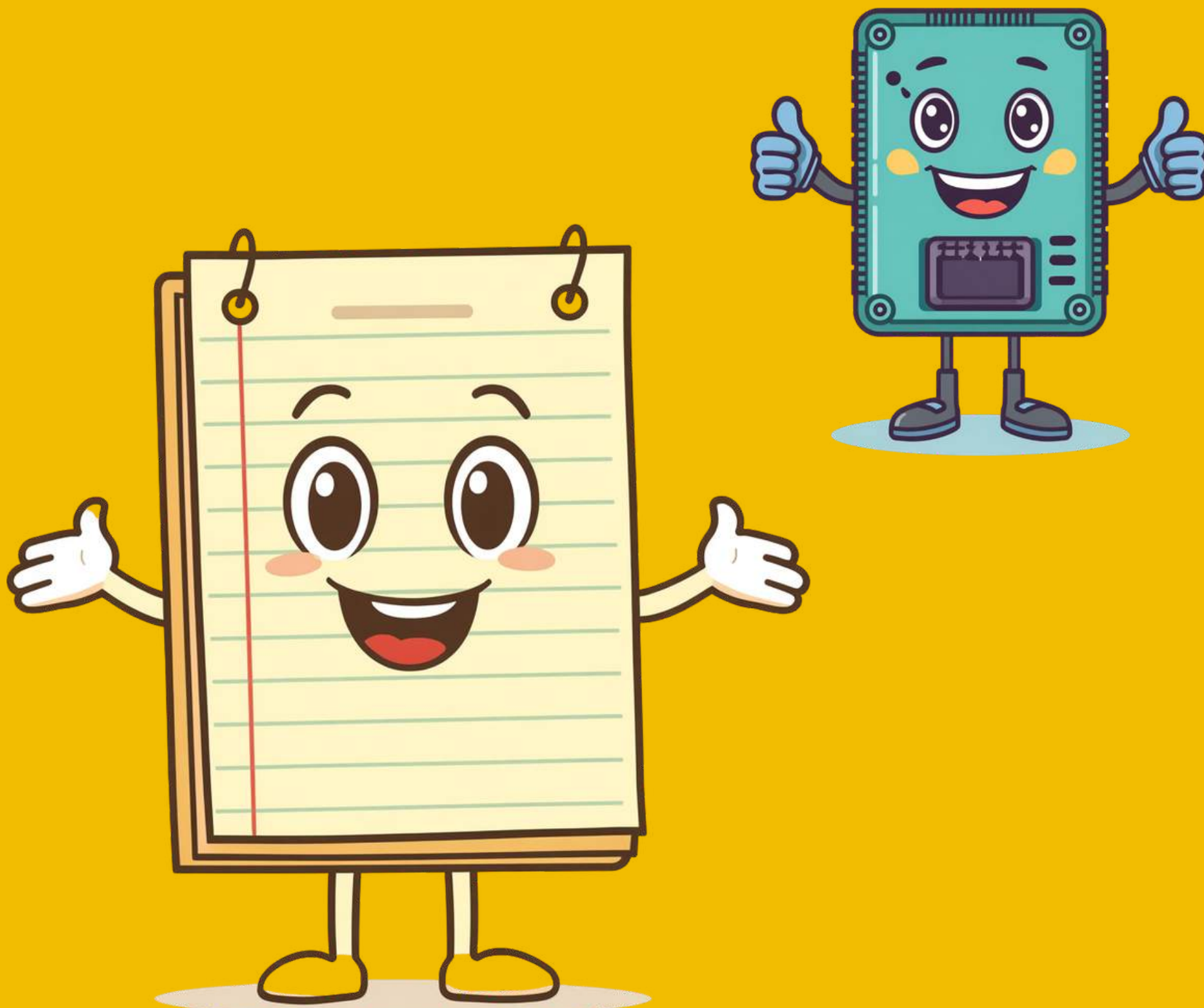
THE TRANSLATOR

The L298N Motor Driver is the power booster — like a railway signalman at CSMT directing trains safely!

QUICK CHECK!

Can Arduino power big motors alone? No! Why? Motors need more current than Arduino can safely provide.

WHY L298N?



DIRECT CONNECTION RISK

Connecting a motor directly to Arduino can damage the board — motors need more current than Arduino can safely provide!

L298N TO THE RESCUE

L298N acts like a railway signalman at CSMT — it safely controls motor direction and boosts power from an external battery.

L298N PINS

Key pins: IN1, IN2 (direction signals from Arduino), OUT1, OUT2 (to motor), VCC (battery power), GND (common ground).

QUICK QUIZ!

Which pins tell the motor which way to spin? Hint: Look at the IN pins! IN1 & IN2 control the direction!

HARDWARE & SAFETY!



POWER SOURCE

Use a 9V or 12V battery to power your motor through the L298N driver — never from Arduino directly!

SAFETY RULE

NEVER connect a motor straight to Arduino pins — it can damage your board permanently!

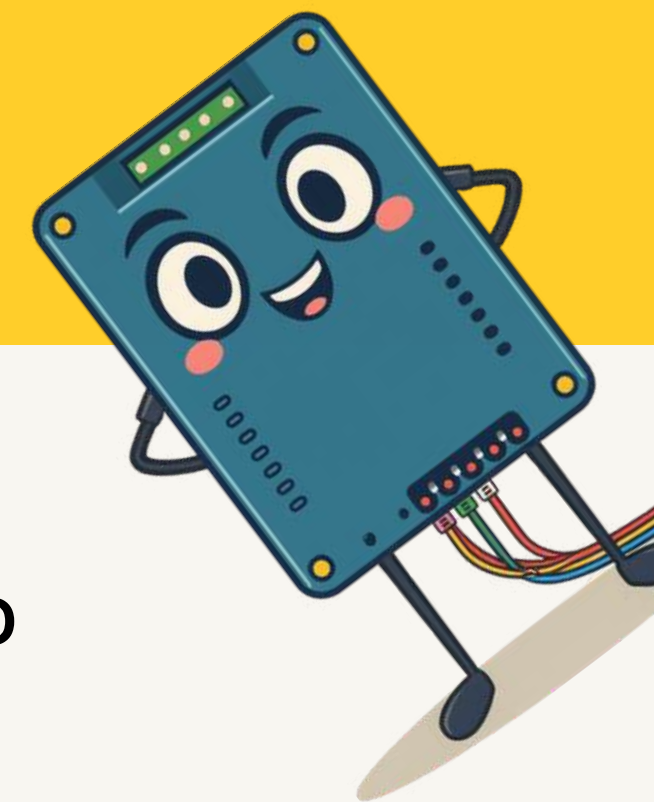
CONNECTIONS

Arduino pins 8 & 9 → L298N IN1 & IN2. Motor wires → OUT1 & OUT2. Arduino GND → L298N GND.

THINK & CHECK!

Why do we connect Arduino GND to L298N GND? So both boards share the same reference voltage — without it, signals won't work!

LET'S BUILD THE CIRCUIT!



Arduino Pin 8 → L298N IN1



Arduino Pin 9 → L298N IN2



Motor Terminals → OUT1 & OUT2



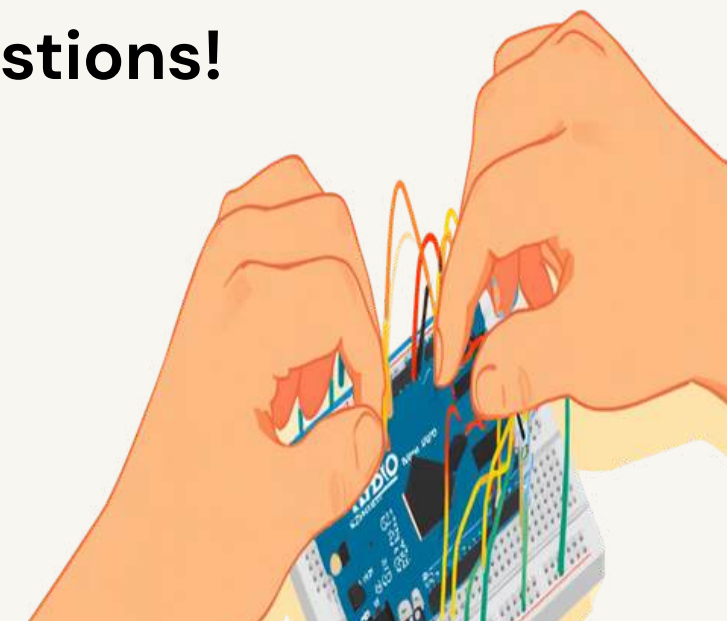
Arduino GND → L298N GND



Battery → L298N VCC & GND

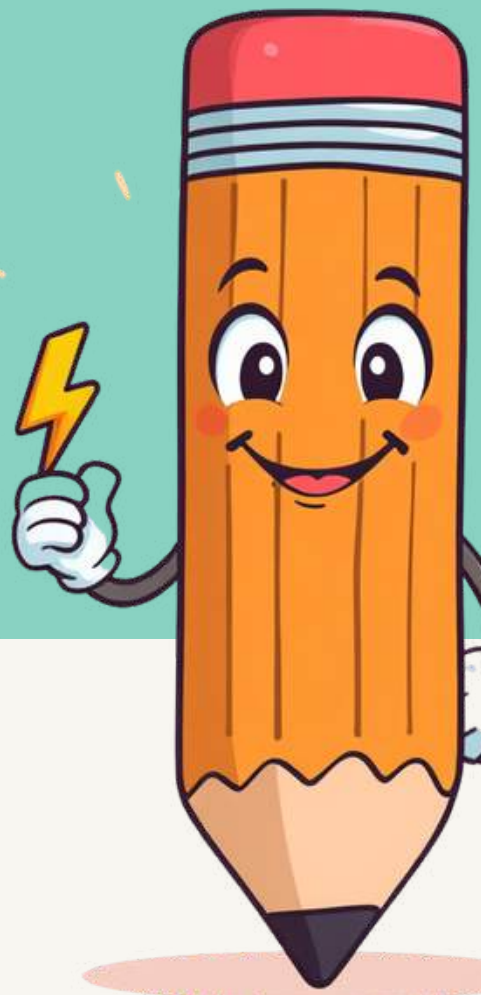


Follow along and ask questions!





CODING TO SPIN THE MOTOR!



1

Forward:
`digitalWrite(8,HIGH);`
`digitalWrite(9,LOW);`

2

Stop: `digitalWrite(8,LOW);`
`digitalWrite(9,LOW);`

3

Backward:
`digitalWrite(8,LOW);`
`digitalWrite(9,HIGH);`

4

Use `delay()` for timing: 2 sec
forward, 1 sec stop, 2 sec
backward!

5

Like telling a ceiling fan
which way to spin!

6

What if both pins are
HIGH?

FUN CHALLENGE!



THE PATTERN

1 second clockwise, then 1 second counter-clockwise — repeat like a monsoon wiper!

TROUBLESHOOTING

Check all GND connections are properly joined between Arduino and L298N.

TROUBLESHOOTING

Use correct delay() times in your code so the motor switches direction smoothly.

TRY IT YOURSELF!

Modify the delay values and watch how the motor behaves — like a wiper in heavy Mumbai monsoon rain!

RECAP & QUIZ TIME!



Arduino is the Brain



L298N Controls Motor Safely



Wiring: IN1, IN2, OUT1, OUT2, GND, VCC



Code Controls Direction & Timing



Quiz: IN1=HIGH & IN2=HIGH → Motor Stops!



Quiz: Why connect Arduino GND to L298N GND?



**THANK YOU & SEE YOU NEXT
TIME!**

